

The Gogol Effect in LLMs:

Post-Completion Self-Devaluation

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Abstract

Prior work shows that LLMs can favor content attributed to themselves, suggesting an IKEA-like self-bias. We test whether this explains why models evaluate their own text differently during generation and after completion. A Live–Replay–Post protocol separates an IKEA-like writer-state shift (Replay–Live) from post-completion self-devaluation (Post–Replay). Across six models with Live–Replay–Post data, only one has a positive mean IKEA-like shift (+2.8 points); the other five shift in the opposite direction. By contrast, four of six show positive evidence of post-completion self-devaluation, with mean shifts from +3.2 to +20.1 points; the remaining two are near zero (+0.2) or negative (−2.0). In the strongest case, 88% of the total Live–Post change emerges only after completion. We call this model-dependent pattern the *Gogol effect*: a model judges its own earlier text more harshly once it can inspect the finished output.

1 Introduction

Large language models increasingly evaluate their own intermediate outputs. Agents critique actions before execution, reasoning systems score partial trajectories, and iterative generation methods use self-evaluation to decide whether to continue, revise, or abandon an answer. Yet the score produced *during* generation is not necessarily the score the same model would assign after the output is complete. This matters whenever an online controller treats a live self-score as if it were a stable estimate of the model’s eventual judgment.

One plausible explanation is an LLM analogue of the human *IKEA effect*, in which people can value products more after contributing labor to them (Norton et al., 2012). Recent LLM work makes that analogy plausible: models can favor outputs associated with themselves or their model family, exhibit self-attribution bias when an action is implicitly framed as their own, and report greater confidence in identical answers when those answers appear as their own assistant response (Panickssery et al., 2024; Xu et al., 2024; Khullar et al., 2026; Sanz-Guerrero et al., 2026). If this is what drives live-versus-retrospective disagreement, then a model should become less favorable to the same partial output when moved from the original writing trajectory into a fresh observer-like evaluation.

But there is another possibility. A sentence that looked acceptable when written may look repetitive, formulaic, inconsistent, or otherwise weak once the entire response is visible. In that case, the main shift should occur not when the model leaves the writing trajectory, but only after the completed work can be inspected as a whole. We call this second pattern *post-completion self-devaluation*. For memorability, we refer to it as the *Gogol effect*: Nikolai Gogol labored for years on the second part of *Dead Souls* and famously burned most of the manuscript shortly before his death (Encyclopedia of Russian History, 2004). The name is a literary metaphor, not a claim that language models experience regret, ownership, or any

human mental state.

We distinguish the two patterns with a three-state protocol. LIVE records the score emitted on the original generation trajectory. REPLAY reconstructs the visible prefix through the same sentence in a fresh call. POST asks for a retrospective score after the complete output is visible. For sentence i in output q , let the corresponding scores be L_{qi} , R_{qi} , and P_{qi} . Then

$$P - L = (R - L) + (P - R). \quad (1)$$

The identity is exact, although its causal interpretation is not. On our scale, higher scores mean “more likely to appear AI-generated,” so a positive $R - L$ means that Live was more favorable than Replay and is therefore directionally consistent with an IKEA-like writer/self-attribution effect. A positive $P - R$ means that the model judges the earlier text more harshly after seeing the completed output; this is the operational signature we call the Gogol effect.

Our probe asks models to generate short social-network posts and, after each sentence, score from 0 to 100 how likely an average reader would be to regard that sentence as AI-generated. The endpoint is intentionally easy to elicit both online and retrospectively, but it is not a validated probability of human detection and it is not factual-confidence calibration. The experiment therefore studies *state dependence in self-evaluation*, not general correctness overconfidence.

The central empirical result becomes clearer when Replay is available across model families. Among six models with Live–Replay–Post data, only Gemini has a positive mean Replay–Live shift (+2.77 points); the other five means are negative, ranging from −3.01 to −8.40. Post–Replay behaves differently. GPT-5.5, Gemini 3.1 Pro, GPT-OSS-120B, and DeepSeek V4 Flash show positive post-completion shifts of +6.91, +20.06, +10.00, and +3.18 points, respectively, whereas Llama 3.3 70B is near zero (+0.24) and Claude Opus 5 trends in the opposite direction (−2.05). Thus IKEA-like self-favoring is uncommon and modest in this sample, while the

Gogol effect appears in more models and can be substantially larger. The effect is nevertheless model-dependent rather than universal.

Contributions.

- We distinguish two phenomena that a simple Live–Post comparison conflates: an IKEA-like writer/self-attribution component ($R - L$) and post-completion self-devaluation ($P - R$), which we call the Gogol effect.
- We introduce a matched Live–Replay–Post protocol that operationally decomposes the retrospective shift while making explicit the limits of hosted-model replay as a causal intervention.
- Across six decomposed models, only one has a positive mean IKEA-like shift, whereas four show positive evidence of a Gogol effect. Among those four, mean Post–Replay shifts range from +3.18 to +20.06 points; two additional models show no positive effect at the available sample sizes.
- A cost-controlled adaptive Replay extension broadens the decomposition beyond the two original confirmation models. Fresh-extension sequential tests support the positive Gogol effect in GPT-OSS-120B and DeepSeek V4 Flash, while Claude Opus 5 and Llama 3.3 70B remain negative or near zero through 24 prompts.
- We show that score-level shifts and within-output ranking stability are separable, which matters for systems that use self-scores to choose which span to revise rather than merely whether a threshold has been crossed.

2 Related Work

Confidence during generation. LLMs can provide informative self-estimates of correctness under some elicitation schemes (Kadavath et al., 2022), and self-generated feedback is used by iterative methods such as Self-Refine and Reflexion (Madaan et al., 2023; Shinn et al., 2023). More recent work makes confidence explicitly trajectory-dependent. FineCE estimates confidence at intermediate points and uses later text to revise confidence about earlier prefixes (Han et al., 2025); premature-confidence work shows that early commitment can predict flawed reasoning (Gai et al., 2026); and trajectory-grounding work shows that apparently calibrated confidence can remain insufficiently sensitive to the actual reasoning trajectory (Yang et al., 2026). These results rule out any broad novelty claim that confidence has only been studied post hoc. Our contribution is the matched comparison of native live scoring, prefix replay, and completed-output retrospective scoring for the same generated material.

Self-preference, self-attribution, and ownership. LLM judges exhibit systematic evaluator biases despite often correlating with human preferences (Zheng et al., 2023; Wang et al., 2024). Models can favor outputs from themselves or their model family (Liu et al., 2024; Panickssery et al., 2024; Wataoka et al.,

2024), and self-refinement can amplify self-bias (Xu et al., 2024). Most directly, Khullar et al. (2026) show that identical actions may be judged more favorably when conversational structure implicitly attributes them to the monitor itself, while Sanz-Guerrero et al. (2026) show higher confidence for identical answers presented as the model’s own assistant response rather than as user content. We therefore do not claim to discover self-favoring in LLMs. Instead, we ask whether such known effects account for the much larger difference that can appear between evaluation during generation and evaluation after completion.

From IKEA-like self-favoring to the Gogol effect. The human IKEA effect describes increased valuation associated with one’s own labor (Norton et al., 2012). We use it only as a directional analogy: a positive Replay–Live shift means the model was more favorable to the text on its original generation trajectory than when the same visible prefix was reconstructed in a fresh call. This is compatible with self-attribution or ownership-like bias but is not a clean causal estimate of either construct. The second pattern we emphasize occurs later: if Post–Replay is positive, the model becomes more critical only after the output is complete. We call this post-completion self-devaluation the *Gogol effect*. The term is deliberately metaphorical and does not imply human-like regret or dissatisfaction.

Self-correction and inference-time control. Self-evaluation is operationally important because it is consumed by selectors, verifiers, and correction systems (Wang et al., 2023; Lightman et al., 2023; Snell et al., 2024). Models may also fail to correct their own errors even when they can correct externally presented ones (Tsui, 2025), while trained approaches such as SCoRe target self-correction directly (Kumar et al., 2025). Our study does not evaluate a general self-correction capability; it asks when a live self-score can be treated as the same ordering or threshold signal that would be available later.

3 Operationalizing IKEA-Like and Gogol Effects

3.1 Three Evaluation States and Two Candidate Effects

Let L_{mqi} denote model m ’s Live score for sentence i in output q , R_{mqi} the Replay score, and P_{mqi} the Post score. Higher values mean “more likely to appear AI-generated.” The goal is not to estimate a human probability; the goal is to compare the same scoring rule across model states.

We use two signed contrasts throughout. Because lower scores are more favorable to the generated text, an *IKEA-like shift* is $R - L > 0$: the model scores its output more favorably while it is still on the writing trajectory than when the same visible prefix is replayed in a fresh context. The *Gogol effect* is $P - R > 0$: the model scores that earlier text less favorably after the completed output becomes available. These are op-

erational definitions tied to the present elicitation; neither is asserted to be a psychological mechanism.

Rank reliability. For local selection, the key quantity is whether Live preserves the ordering later assigned by Post. We report the mean within-output Spearman correlation

$$\mathcal{R}_m = \frac{1}{|Q_m|} \sum_{q \in Q_m} \rho_s(L_{mq}, P_{mq}), \quad (2)$$

and pairwise ordering agreement among non-tied sentence pairs. These metrics directly match decisions such as “which span should be revised?” and avoid conflating across-prompt scale differences with local ordering.

Level agreement. We report bias $B_m = \mathbb{E}[L - P]$ and mean absolute error $E_m = \mathbb{E}|L - P|$. A stable level shift may be correctable without changing the ranking. We therefore fit model-specific Live→Post mappings under five-fold cross-validation grouped by prompt, comparing raw Live scores with an offset-only and an affine correction.

State stability. Replay tests whether a fresh call given approximately the same textual prefix reproduces the Live judgment. Equation 1 separates the total retrospective shift into an operational Replay–Live component and a Post–Replay component. The latter includes the effect of seeing later sentences and any other difference induced by the retrospective prompt. The former includes the difference between being on the original generation trajectory and reconstructing the prefix in a fresh call, plus any divergence in earlier score-token history.

These axes define a simple taxonomy. A live score may be (i) rank-stable and level-aligned, (ii) rank-stable but level-shifted and therefore potentially calibratable, or (iii) rank-unstable, in which case a monotone level correction cannot make it a reliable local selector.

3.2 What the Framework Does and Does Not Identify

The protocol measures *state-indexed consistency*, not evaluator truth. Post is a retrospective same-model comparator, not human ground truth or an external reward model. Consequently, weak Live–Post agreement can arise because Live is state-sensitive, because Post is a poor evaluator, or both. The intended use is diagnostic: if a controller will later rely on a retrospective or delayed score, the audit tests whether its online surrogate preserves the required ordering and level. Establishing external validity requires a separate task-specific judge benchmark.

Likewise, Replay is not a hidden-state intervention. Hosted APIs do not expose the internal state needed to hold every latent variable fixed, and Replay’s preceding score tokens can diverge from Live. The decomposition is therefore useful for ruling out overly simple interpretations of $P - L$, not for claiming a complete causal explanation.

4 Experimental Design

4.1 Task Probe and Frozen Prompt Sets

We froze a bank of 110 factual social-writing prompts: 100 English primary prompts and 10 non-English validation prompts. Each asks for an approximately 5–8 sentence post suitable for a general social-network audience rather than an encyclopedia entry. Topics span ten domains: arts/culture, biology/medicine, economics, food/culture, geography, history, language, misconceptions, science, and technology. The main English set contains 40 prompts, exactly four from each domain, selected before the adaptive run from the frozen bank.

The scoring criterion is intentionally subjective but fixed: after every sentence, the generating model estimates on a 0–100 scale how likely an average human reader would be to think the sentence was AI-generated. The phrase “average human reader” is part of the elicitation. No measured human judgments enter the reported endpoint, and the paper makes no claim that the score is calibrated to actual human perception.

4.2 Models and Execution

We attempted ten models through fixed route identifiers: GPT-5.5, Claude Opus 5, Gemini 3.1 Pro Preview, DeepSeek V4 Flash, DeepSeek V3.2, GPT-OSS-120B, Llama 3.3-70B-Instruct, Qwen3-30B-A3B-Instruct, Ministral 14B, and Gemma 3-12B-IT. The frozen manifest records the exact route strings. The main run used OpenRouter automatic provider routing with fallbacks enabled; this provider indirection is a reproducibility limitation and is reported rather than hidden. All primary Live, Post, Replay, and correction calls used temperature 0.7, including numeric-score calls. Exact templates, caps, seed, route identifiers, and parsing rules are in Appendix A.

4.3 Live, Replay, and Post

In LIVE, the model generates the requested post and appends a machine-parseable integer score after every sentence. The ordinary punctuation appears before the score tag. The trajectory is then parsed into clean text and sentence scores.

In POST, a fresh call receives the completed clean output with one score placeholder per sentence and returns retrospective sentence scores. Because Post sees the completed output, it has access to future sentences unavailable to earlier Live scores.

In REPLAY, a fresh call reconstructs the sequential textual prefix through the target sentence and requests only that sentence’s score. Replay proceeds sentence $1 \rightarrow N$; later Replay steps contain Replay’s own earlier scores rather than the original Live scores. The initial experiment replayed a preselected 10% subset; Replay was later expanded to all 40 English prompts for GPT-5.5 and Gemini 3.1 Pro and all 10 non-English validation prompts.

4.4 Cross-Model Replay Breadth Extension

The main study expanded Replay to all 40 English prompts for GPT-5.5 and Gemini 3.1 Pro. To test whether the decomposition generalized beyond those two models, we subsequently extended Replay for four additional models selected for two operational reasons fixed before the extension: their original Live/Post cells provided sufficient usable English coverage, and the existing Replay parser produced valid single-score responses. The breadth cohort is Claude Opus 5, DeepSeek V4 Flash, GPT-OSS-120B, and Llama 3.3-70B-Instruct. Qwen3 and Gemma 3 were not included because the earlier Replay calls frequently violated the single-score output contract; DeepSeek V3.2 and Ministral had substantially lower usable Live/Post coverage.

An already-inspected four-prompt Replay pilot was retained in effect-size estimates rather than discarded. We then froze a cost-saving adaptive extension in increments of four additional prompts, with a maximum total of 24 prompts per model. Models stopped receiving API calls once the exploratory cumulative one-sided test for $P - R > 0$ crossed .05; unresolved models continued to the next increment. Because the stopping decision was chosen after the initial four-prompt pilot had been inspected, cumulative stopping p -values are descriptive rather than pristine confirmatory tests. For inferential support we additionally test the *new* extension prompts only and apply a Bonferroni correction over the five possible future looks. Under this sequentially adjusted fresh-extension test, GPT-OSS-120B crosses at total $N = 8$ ($p_{\text{adj}} = .021$) and DeepSeek V4 Flash at $N = 12$ ($p_{\text{adj}} = .040$). Claude and Llama continue to the 24-prompt cap without positive evidence. The frozen prompt order, all interim looks, and both cumulative and sequential statistics are included in the reproduction package.

4.5 Pre-Specified Adaptive Confirmation

The original confirmatory design used planned looks at $N \in \{40, 50, 60, 70, 80, 90, 100\}$ and familywise $\alpha = .05$, corresponding to a per-look Bonferroni boundary of .00714. Test 1 was a prompt-level Spearman correlation between mean Live and mean Post scores, tested against $\rho_0 = .2$. Tests 2–3 required Gemini’s mean Post–Live shift to exceed 5 points and the matched Gemini-minus-GPT-5.5 shift difference to exceed 5 points. All three had to cross the boundary at the same planned look; the run stopped at $N = 40$. Pilot data were excluded.

The within-output ranking analysis, Replay decomposition, grouped calibration, and correction analyses were specified after the main generations were frozen and are reported as secondary analyses. This distinction is important because the paper’s principal methodological contribution does not depend on treating those endpoints as preregistered.

4.6 Held-Out Robustness Studies

After the main experiment was frozen, we tested whether the observed state dependence was an artifact of the exact scoring

surface. A first frozen set of 16 new prompts was evaluated with GPT-5.5 and Gemini 3.1 Pro under four semantically matched elicitation schemes: the original AI-likeness score/tag; a semantic paraphrase; a reverse-coded human-likeness score transformed as $100 - n$; and the original semantics with a minimal numeric tag. Live and Post were collected for every cell, with Replay on a preselected subset.

After inspecting that cohort, we froze a separate set of 24 new prompts and repeated Live and Post under the same two models and four elicitation schemes. This extension is explicitly a follow-up confirmation rather than a preregistered replication of the first cohort. We also froze a separate 10-prompt temperature control and ran full Live–Replay–Post for GPT-5.5 and Gemini at both $T = 0$ and $T = 0.7$.

4.7 Equal-Compute Rewrite Check

For GPT-5.5 and Gemini, we tested one downstream policy. The guided arm rewrites the sentence with the highest Live score; a paired random arm rewrites a deterministic random different sentence with the identical instruction. The model receives the original task, the full frozen post, and the target sentence, and returns one replacement sentence. After splicing it into the otherwise frozen post, a fresh Post pass scores the whole corrected document. This experiment is a policy check, not a general test of self-correction capability.

5 Results

5.1 Live and Retrospective Self-Evaluation Can Differ Substantially

The original adaptive confirmation stopped at the first planned look ($N = 40$) because all three pre-specified tests crossed the sequential boundary. GPT-5.5 showed positive prompt-level Live–Post association ($\rho = .599$, one-sided $p = .00147$ against $\rho_0 = .2$). Gemini’s mean Post–Live shift was +22.83 points ($p = 6.9 \times 10^{-7}$), exceeding GPT-5.5’s shift by 19.02 points ($p = .00111$). Thus a live score can contain information about the model’s later judgment while still changing substantially in level.

A two-state result like this is compatible with an IKEA-like interpretation, but does not identify it. The Replay condition is what distinguishes the writer-state-associated part of the discrepancy from the additional change that appears when the completed output becomes available.

5.2 IKEA-Like Self-Favoring Is Uncommon in the Six-Model Decomposition

The six-model decomposition provides little support for a general IKEA-like writer-state effect. Gemini is the only model with a positive mean Replay–Live shift: +2.77 points on 40 English prompts (one-sided t test $p = .010$; one-sided Wilcoxon $p = .029$). Its magnitude is modest relative to its later Post–Replay shift. The other five models all have negative mean Replay–Live shifts: GPT-5.5 -3.10 , GPT-OSS-120B -6.48 ,

Table 1: Live–Replay–Post decomposition across six models. $R - L > 0$ is IKEA-like; $P - R > 0$ is the Gogol direction. N is the number of prompts with complete decomposition used for that model. The four breadth-extension sample sizes are adaptive; see text.

Model	N	$R - L$	$P - R$	Summary
GPT-5.5	40	−3.10	+6.91	Gogol
Gemini 3.1 Pro	40	+2.77	+20.06	IKEA-like + Gogol
GPT-OSS-120B	8	−6.48	+10.00	Gogol
DeepSeek V4 Flash	12	−8.40	+3.18	Gogol
Llama 3.3 70B	24	−3.81	+0.24	no Gogol evidence
Claude Opus 5	24	−3.01	−2.05	opposite/none

DeepSeek V4 Flash −8.40, Llama 3.3 70B −3.81, and Claude Opus 5 −3.01 points (Table 1). Thus the dominant cross-model pattern is not systematic self-favoring while writing.

This result does not contradict prior demonstrations of ownership or self-attribution bias. Replay is a specific transition from the original generation trajectory to a reconstructed prefix, not a direct ownership manipulation. The result is narrower: in this probe, the writer-to-Replay transition rarely moves in the IKEA-like direction, and when it does the positive shift is small compared with the strongest post-completion changes.

5.3 The Gogol Effect Is More Frequent and Often Much Larger

Post–Replay shows a different pattern. GPT-5.5 and Gemini have large positive shifts of +6.91 and +20.06 points on the fixed 40-prompt confirmation set (one-sided $p = 7.8 \times 10^{-14}$ and 5.8×10^{-11}). In Gemini, $20.06/22.83 \approx 88\%$ of the total Live–Post change appears only after the completed output is available.

The breadth extension adds two further positive cases. GPT-OSS-120B reaches a cumulative mean Post–Replay shift of +10.00 points at total $N = 8$; the four fresh extension prompts independently cross the pre-frozen sequential correction ($p_{\text{adj}} = .021$). DeepSeek V4 Flash reaches +3.18 points at total $N = 12$, with fresh-extension sequential $p_{\text{adj}} = .040$. These effects coexist with strongly negative Replay–Live means, so they cannot be described as simple persistence of an IKEA-like writer-state shift.

The two negative cases are informative. Llama 3.3 70B remains near zero at the 24-prompt cap (+0.24; cumulative one-sided $p = .455$, descriptive 95% interval $[-4.10, 4.58]$). Claude Opus 5 moves in the opposite direction (−2.05; $p = .972$, interval $[-4.15, 0.06]$). We therefore do not claim that the Gogol effect is universal. Rather, it is a recurrent, sometimes large, model-dependent form of state instability: positive evidence appears in four of the six models for which the Live–Replay–Post decomposition is sufficiently developed, compared with a positive IKEA-like mean in only one.

The original controls continue to support the interpretation in the two high-coverage confirmation models. In the 16-prompt Replay robustness cohort, Post–Replay exceeds Replay–Live in all eight model–elicitation combinations. In the separate $T = 0$ control, Replay–Live is −1.97 and −0.01, while

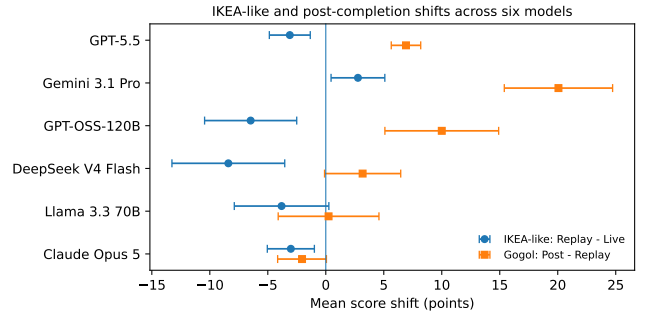


Figure 1: Mean IKEA-like ($R - L$) and Gogol ($P - R$) shifts across six models. Error bars are descriptive 95% t intervals over prompts. GPT-5.5 and Gemini use fixed $N = 40$ samples; the other four use adaptive breadth-extension sample sizes, so their intervals should not be interpreted as fixed-design confidence statements.

Post–Replay is +7.13 and +25.27 for GPT-5.5 and Gemini. Thus the post-completion shift in those models does not require stochastic sampling of the numeric score token at $T = 0.7$.

5.4 Retrospective Level Shifts and Rank Reliability Are Different Problems

The Live–Post gap does not fully characterize whether a live self-score is useful. Across the ten-model descriptive sweep, mean within-output Live–Post Spearman correlation ranges from .329 to .820, while raw mean absolute error ranges from 4.2 to 24.5 points (full table in Appendix C). These axes do not move together. One model in the sweep has the smallest raw MAE but among the weakest local rankings; another has a large level shift and weak ranking; several preserve ordering despite nonzero bias.

This distinction matters operationally. A monotone calibration can move a threshold but cannot reorder sentences. Five-fold prompt-grouped cross-validation reduces held-out MAE for several models, including a reduction from 24.47 to 20.07 points in the largest-shift case, but such calibration leaves the within-output ranking unchanged. A system choosing *which* span to revise therefore needs a rank audit, not merely a calibration curve.

5.5 Held-Out Elicitation Preserves the Direction of the Retrospective Shift

A separate 24-prompt follow-up repeated Live and Post under four semantically matched scoring surfaces. All eight model–elicitation mean Post–Live shifts were positive with 95% prompt-bootstrap intervals above zero. Averaged across variants within each fresh prompt, GPT-5.5 shifts +5.33 points and Gemini +8.61, with a matched difference of +3.28 (95% CI $[0.72, 5.97]$). The magnitude of within-output rank reliability is more prompt-sensitive, reinforcing the distinction between a directional level effect and ranking stability.

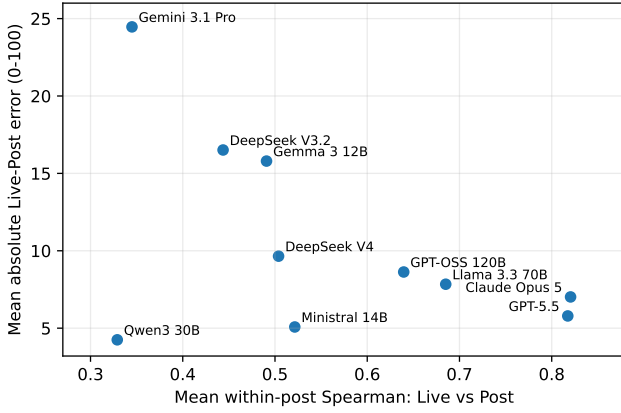


Figure 2: Rank reliability and level error are separable properties of live self-evaluation. The plot is diagnostic rather than a model leaderboard; exact values are reported in Appendix C.

5.6 A Downstream Check: Good Target Ranking Is Not Sufficient for One-Shot Rewriting

We also tested whether choosing the highest Live-scored sentence improves a fixed one-sentence rewrite relative to an equal-compute random target. It does not reliably do so. On GPT-5.5 ($N = 40$), the mean guided advantage is -0.28 Post-score points; on Gemini, 37 valid pairs yield $+0.27$ points, with neither win-rate test significant. This null result is particularly informative for GPT-5.5 because its Live target is retrospectively the highest-scored sentence in 72.5% of outputs and in the top two in 87.5%. Accurate target identification is therefore not sufficient for this correction operator to improve the document. This is a policy check, not a general limit on self-correction.

6 Discussion

6.1 The Main Finding: IKEA-Like Bias Is Rare; the Gogol Effect Is More Frequent

The six-model decomposition does not support a simple story in which LLMs generally overvalue their own work while producing it. Only one model has a positive mean Replay–Live shift, and that shift is modest. Five move in the opposite direction. This is compatible with prior work showing ownership and self-attribution bias because our Replay manipulation is not identical to those studies; it does show that such biases do not generally explain the live-to-retrospective gap in this probe.

Post-completion self-devaluation is more frequent in the present sample. Four of six models show positive evidence of a Gogol effect, and the positive mean shifts span roughly $+3$ to $+20$ points. Two do not: Llama is approximately unchanged, while Claude trends toward post-completion *upvaluation*. The heterogeneity is part of the result. It argues against treating either “IKEA” or “Gogol” as a universal anthropomorphic trait and instead suggests that self-evaluation dynamics are properties of particular model/training/serving combinations.

The two contrasts also occur at different transitions. A

model can show both, as Gemini does: a modest IKEA-like $R - L > 0$ shift followed by a much larger Gogol $P - R > 0$ shift. Other models, including GPT-OSS and DeepSeek V4, move in the opposite direction at Replay but then reverse toward strong post-completion devaluation. A two-state Live–Post design would hide these qualitatively different trajectories.

6.2 What the Gogol Effect Does and Does Not Mean

The Gogol effect is an operational pattern, not evidence that an LLM experiences remorse. The Post condition has information unavailable to the corresponding Live or Replay state. Seeing later sentences may reveal repetition, stylistic uniformity, contradiction, or an overall structure that changes the interpretation of an earlier sentence. The retrospective prompt also changes the evaluation context. Our claim is therefore narrower: for some models, the dominant live-to-retrospective shift is associated with *completion*, not with the writer-versus-observer contrast measured by Replay.

This distinction also prevents an overreading of a two-state Live–Post experiment. A large positive Post–Live difference can look like “the model liked its work while making it.” The Replay condition shows that the same observation can instead be mostly “the model judged an unfinished artifact without evidence that became available later.” Those explanations have different implications for online control.

6.3 Relation to Existing Trajectory-Confidence Work

Our findings are compatible with recent evidence that confidence is not a static property of an answer. FineCE explicitly uses future text to improve confidence estimates for earlier prefixes (Han et al., 2025); premature-confidence work studies how confidence evolves during reasoning (Gai et al., 2026); and trajectory-grounding work asks whether verbalized confidence responds to the contents of a reasoning trajectory (Yang et al., 2026). We therefore do not claim that later context influencing confidence or evaluation is itself novel.

The narrower contribution is the matched decomposition of an apparent self-favoring gap. Existing ownership/self-attribution studies manipulate who appears to own otherwise identical content. Existing trajectory-confidence studies examine confidence through a developing reasoning process. Live–Replay–Post combines these perspectives to ask how much of an online-to-retrospective shift survives when visible prefix context is held approximately fixed, and how much appears only after completion. Across the six-model decomposition, the completion-associated component is positive in four models and dominates the strongest observed shifts, while two models show little or no positive completion effect.

6.4 Implications for Self-Evaluation and Inference-Time Control

The practical implication is not that retrospective evaluation is always better. Post has more information, but it is unavailable to an online controller at the moment of decision. Instead, live self-scores should be audited for the property a controller actually uses. Thresholding requires level stability or validated recalibration. Local selection requires ordering stability. A score whose ranking changes after completion cannot be repaired by adding or multiplying a constant.

The Gogol effect also suggests caution when interpreting apparent “overconfidence while generating.” A live model may indeed be self-favoring, but a large Live–Post difference can also be produced by missing future context. Designs that omit a prefix-matched Replay condition conflate these explanations.

7 Limitations

The empirical probe is perceived AI-likeness in short social-writing outputs, not objective correctness. The scoring instruction references an “average human reader,” but no human ground-truth judgments are used; the study therefore establishes state-indexed agreement among model scores, not human-valid detection ability. The motivating examples of premature confidence and building on earlier mistakes concern correctness, whereas our measured endpoint concerns stylistic self-evaluation. Generalizing the Gogol effect to correctness confidence requires a separate experiment.

Replay fixes the visible prose prefix but cannot fix hidden activations, provider state, or the exact earlier score-token sequence. Equation 1 is algebraically exact, while causal labels for its components are necessarily approximate. A positive Replay–Live shift is compatible with an IKEA-like or ownership-like explanation but is not a clean causal estimate of either construct. Likewise, Post–Replay includes both additional completion information and prompt/state differences associated with retrospective evaluation.

Only six models have sufficiently developed Live–Replay–Post data for the cross-model effect comparison, and the breadth-extension sample sizes differ because models were stopped adaptively to limit API cost. The initial four-prompt pilot for those additional models had already been inspected before the extension was designed. We therefore distinguish exploratory cumulative stopping statistics from sequentially adjusted tests on the fresh extension prompts; the latter support the two positive breadth replications reported here. The resulting 4-of-6 versus 1-of-6 contrast is an empirical description of this model sample, not an estimate of prevalence over all LLMs.

Coverage is uneven for several other models because strict-format failures are missing rather than imputed; low-coverage estimates may exhibit survivorship bias. The 24-prompt elicitation robustness extension was frozen after the initial 16-prompt cohort had been inspected, so it is a sequential follow-up rather than an independently preregistered replication. The temperature control contains only 10 prompts. The one-shot rewrite test covers one fixed correction operator. Finally, commer-

cial model routes and hosted serving infrastructure can change over time; the durable claims of the paper concern the Live–Replay–Post distinction and the existence of substantial model-dependent post-completion self-devaluation, not the relative ranking of named model versions.

8 Conclusion

Across six models with Live–Replay–Post decomposition, IKEA-like self-favoring is uncommon: only one model has a positive mean Replay–Live shift (+2.77 points), while the other five means are negative. Post-completion self-devaluation is more frequent and often substantially larger. Four models show positive evidence of a Gogol effect, with mean Post–Replay shifts of +3.18 to +20.06 points; one additional model is near zero and one shifts in the opposite direction. In the strongest case, about 88% of the total Live–Post shift emerges only after the complete output is visible.

We call this model-dependent pattern the *Gogol effect*: an earlier span is judged more harshly once the model can inspect the finished response. The label is metaphorical, but the operational lesson is concrete. A large discrepancy between online and retrospective self-scores need not be an IKEA-like ownership bias; it may emerge only after completion, and whether that happens depends strongly on the model. Systems that consume self-scores during generation should therefore audit the temporal stability of those scores rather than assuming that a live evaluation is interchangeable with a retrospective one.

Reproducibility Statement

The frozen experimental package used to produce the reported numbers contains the prompt bank, manifests, route identifiers, SQLite workspaces, exported score tables, replay and correction artifacts, robustness and temperature-control runs, the adaptive cross-model Replay extension with every interim look, analysis scripts, and expected deterministic analysis outputs. The main run records seed 20260907, prompt-bank version, scoring instruction, planned looks, provider policy, output caps, model route strings, temperature, retry policy, and file hashes. Analyses are reproducible from frozen outputs without provider credentials; exact regeneration of hosted model outputs is not guaranteed because model serving and provider routing are external dependencies.

The public reproduction package is available at <https://github.com/victorlavrenko/llm-gogol-effect>. The repository contains the frozen artifacts, analysis code, adaptive Replay extension, and scripts needed to reproduce the reported analyses from stored outputs. The repository is not required to interpret the statistical claims in this manuscript; all central methods, exclusions, and numerical endpoints are stated here or in the appendices.

Ethics Statement

The quantitative experiments use synthetic writing prompts, model-generated text, and model-produced scores. No private or personally identifying human data contribute to the reported estimates. The “average human reader” phrase is part of the model-scoring instruction and should not be interpreted as a measurement of actual human opinion. The study compares properties of model-generated control signals and does not rank demographic or protected groups.

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A Frozen Prompt and Execution Protocol

The frozen prompt bank contains 100 English primary prompts and 10 non-English validation prompts. The English main subset is balanced at four prompts per domain. The generated-post contract is 5–8 sentences. The scoring scale is 0 (very human-like) to 100 (obviously AI-generated), estimating how likely an average human reader would be to think the sentence was AI-generated. The frozen system instruction is: “Follow the user’s writing task and formatting instructions exactly. Return only the requested output.”

The main API run was executed on September 7, 2026 (UTC timestamps in the SQLite log span 12:28–13:38). All primary conditions used temperature 0.7. Output caps were 4096 tokens for Live and Post and 1024 for each Replay call; no explicit input truncation was applied. Seed was 20260907, with up to 20 global workers, at most one worker per model, and three retries. The frozen bank version is `prompts-v3`; its SHA-256 is recorded in the manifest. Pilot data were excluded.

The main provider policy was “OpenRouter automatic provider routing with fallbacks enabled.” The frozen route identifiers were:

- `openai/gpt-5.5`
- `anthropic/claude-opus-5`
- `google/gemini-3.1-pro-preview`
- `deepseek/deepseek-v4-flash-0731`
- `deepseek/deepseek-v3.2`
- `openai/gpt-oss-120b`
- `meta-llama/llama-3.3-70b-instruct`
- `qwen/qwen3-30b-a3b-instruct-2507`
- `mistralai/ministral-14b-2512`
- `google/gemma-3-12b-it`

Common user prefix. Every Live, Post, and Replay user message begins with the original writing task followed by the same scoring instruction. Live then appends:

```
Produce the requested text now and
follow the scoring instruction
exactly.
```

Post template. After the common prefix, Post receives the complete frozen text with one `<AI SCORE: ?>` placeholder per sentence and the instruction: “The requested text has already been produced below. Assess that exact completed text without rewriting it. There are N placeholders, one for each sentence. Return exactly the same number of `<AI SCORE: n>` tags, one per line and in sentence order. Do not repeat, continue, or rewrite the text.”

Replay template. After the common prefix, Replay receives only the text prefix through the target sentence, with the final score replaced by `<AI SCORE: ?>`. It is instructed: “Continue the assessment for the text below. The only missing content is the single final `<AI SCORE: ?>`. Do not repeat, continue, or rewrite the text. Replace that placeholder with the score required by the scoring instruction and return only `<AI SCORE: n>`.” Later Replay steps include earlier *Replay* scores, not the corresponding Live scores.

B Model Coverage

Table 2: English paired Live–Post coverage used for the reliability table.

Model	Paired prompts
GPT-5.5	40/40
Gemini 3.1 Pro Preview	40/40
GPT-OSS-120B	40/40
Llama 3.3 70B Instruct	40/40
Qwen3 30B A3B Instruct	39/40
Claude Opus 5	37/40
Gemma 3 12B IT	35/40
DeepSeek V4 Flash	32/40
DeepSeek V3.2	18/40
Ministral 14B	12/40

Protocol failures were predominantly failures to obey the exact sentence-plus-score contract, making the cell unparseable for paired sentence analysis. No missing score was imputed. Because adherence may correlate with prompt difficulty or output complexity, low-coverage estimates can suffer survivorship bias and should not be read as controlled head-to-head comparisons.

C Full Multi-Model Reliability Results

Table 3: English Live–Post reliability analysis. Q is the number of prompts with paired outputs. $\bar{\rho}_q$ is mean within-output Spearman; Pair is non-tied pairwise ordering agreement. Bias is Live–Post; MAE is on the 0–100 score scale. The table is descriptive rather than a controlled model leaderboard because protocol adherence and provider routes differ across systems.

Model	Q	$\bar{\rho}_q$	Pair	Bias	MAE
Claude Opus 5	37	.820	.866	+4.1	7.0
GPT-5.5	40	.817	.864	−3.8	5.8
Llama 3.3 70B	40	.685	.802	+4.1	7.8
GPT-OSS 120B	40	.640	.792	−5.9	8.6
Ministral 14B	12	.521	.719	−1.3	5.1
DeepSeek V4	32	.504	.723	+2.1	9.7
Gemma 3 12B	35	.491	.705	+4.9	15.8
DeepSeek V3.2	18	.444	.690	−10.9	16.5
Gemini 3.1 Pro	40	.345	.659	−23.4	24.5
Qwen3 30B	39	.329	.649	+0.7	4.2

D Grouped Calibration Results

Table 4: Five-fold prompt-grouped cross-validation. Values are Live→Post MAE; lower is better. Offset estimates one additive correction on training folds. Affine estimates $aL + b$.

Model	Raw	Offset	Affine
Qwen3 30B	4.25	4.47	3.73
Ministral 14B	5.08	5.64	5.37
GPT-5.5	5.79	5.49	5.49
Claude Opus 5	7.02	6.64	6.32
Llama 3.3 70B	7.84	8.41	7.17
GPT-OSS 120B	8.63	7.47	6.87
DeepSeek V4	9.65	9.86	9.16
DeepSeek V3.2	16.51	14.71	14.01
Gemma 3 12B	15.80	16.55	15.43
Gemini 3.1 Pro	24.47	20.07	20.43

These mappings target agreement with the model’s own Post scores. They do not constitute calibration to human probabilities. The folds are grouped by prompt to prevent sentence-level leakage.

E Expanded Replay Statistics

For the fixed 40-prompt English confirmation set, Gemini’s prompt-level mean Replay–Live component is +2.772 points (one-sided t test $p = .0102$; one-sided Wilcoxon $p = .0292$). GPT-5.5’s mean Replay–Live component is −3.103 points (one-sided test in the positive IKEA direction $p = .9995$; the

opposite-direction one-sided $p = .00050$). Their corresponding Post–Replay means are +20.061 and +6.914 points, with one-sided $p = 5.84 \times 10^{-11}$ and 7.76×10^{-14} .

Table 5: Six-model decomposition. For GPT-5.5 and Gemini, $N = 40$ is fixed. For the four breadth models, N is the adaptive cumulative sample at stopping or cap. Cumulative p_G is a descriptive one-sided t test for $P - R > 0$; for positive breadth stops, p_{seq} is the Bonferroni-adjusted one-sided test on fresh extension prompts only.

Model	N	$R - L$	$P - R$	cumulative p_G	fresh p_{seq}
GPT-5.5	40	−3.10	+6.91	7.8×10^{-14}	—
Gemini 3.1 Pro	40	+2.77	+20.06	5.8×10^{-11}	—
GPT-OSS-120B	8	−6.48	+10.00	.00096	.021
DeepSeek V4 Flash	12	−8.40	+3.18	.028	.040
Llama 3.3 70B	24	−3.81	+0.24	.455	1.000
Claude Opus 5	24	−3.01	−2.05	.972	1.000

The adaptive breadth extension began from an already-inspected four-prompt Replay pilot. New prompts were added in frozen blocks of four. GPT-OSS stopped at total $N = 8$ and DeepSeek V4 at $N = 12$ after the exploratory cumulative statistic crossed .05; their fresh-extension sequentially adjusted p -values are .021 and .040, respectively. Claude and Llama continued to the 24-prompt cap. At that cap, Llama’s mean Gogol effect is +0.240 with descriptive 95% interval $[-4.104, 4.583]$, while Claude’s is −2.048 with interval $[-4.152, 0.055]$. These null/opposite cases are retained as part of the main finding rather than being extended until significance.

The 10-prompt non-English validation set for GPT-5.5 and Gemini shows the same broad asymmetry but is too small for a headline estimate: Gemini moves approximately $14.21 \rightarrow 21.53 \rightarrow 50.93$, while GPT-5.5 moves $19.41 \rightarrow 19.09 \rightarrow 26.32$. We treat this as qualitative cross-language support rather than a powered replication.

F Correction Test Details

For each arm the correction call receives the original writing task, the entire frozen generated text, and the target sentence. The exact English instruction is: “Rewrite only the TARGET SENTENCE so that an average human reader would be less likely to think it was AI-generated. Preserve its intended meaning and factual content, keep it as one sentence, and make it fit naturally into the frozen text. Do not rewrite any other sentence. Return only the replacement sentence, with no score, label, explanation, quotation marks around the whole answer, or additional text.” The replacement is spliced into the frozen document; a fresh Post call then scores every sentence. The arm outcome is the corrected document’s mean Post score. The paired advantage is (random corrected change from baseline) minus (guided corrected change from baseline), so positive values favor guided targeting.

Table 6: English equal-compute one-shot correction test. Positive advantage means the Live-guided target improved the corrected-document Post mean more than a random target.

Model	N	Mean adv.	Wins/non-ties	Binomial p
GPT-5.5	40	-0.282	16/37	.838
Gemini 3.1 Pro	37	+0.270	19/34	.304

The corresponding one-sided Wilcoxon p -values are .677 and .392. Gemini began with 40/40 paired Live/Post prompts; three correction pairs were excluded solely because one rewrite response violated the strict one-line/one-sentence parser: random arm on p035, guided arm on p046, and random arm on p092. No exclusion depended on the resulting score.

G Held-Out Elicitation Robustness Details

The 16-prompt cohort was frozen before generation. Full Live, Post, and preselected Replay cells were then collected for GPT-5.5 and Gemini under four elicitation variants. The separate 24-prompt follow-up was frozen only after the initial cohort had been inspected; all $24 \times 2 \times 4 = 192$ Live cells and all 192 Post cells completed, with no Replay by design. No successfully completed cell was regenerated because of its observed score.

Table 7: Complete robustness summary. $\Delta = P - L$ after normalization to a common AI-likeness direction. ρ_{40} is mean within-output Live-Post Spearman on the pooled 40 prompts.

Elicitation	Initial 16 Δ		Fresh 24 Δ		ρ_{40}	
	GPT-5.5	Gemini	GPT-5.5	Gemini	GPT-5.5	Gemini
Original	5.86	14.33	6.56	8.93	.647	.549
Paraphrase	4.67	9.71	2.95	6.11	.557	.447
Reverse-coded	2.47	4.97	1.21	6.24	.501	.353
Minimal tag	7.22	17.20	10.61	13.16	.357	.332

Reverse-coded human-likeness values are transformed as $100 - n$ before analysis. The extension’s eight model-elicitation mean shifts are all positive with prompt-bootstrap 95% intervals above zero. Averaged across the four variants within each matched prompt, the extension’s Gemini-minus-GPT shift is +3.280 points (95% CI [0.724, 5.970]). The corresponding pooled 40-prompt difference is +4.566 points (95% CI [2.051, 7.269]).

The initial 16-prompt Replay results preserve the main decomposition qualitatively. For GPT-5.5, Replay-Live is -0.22, +1.10, -2.05, +0.32 points across original, paraphrase, reverse-coded, and minimal variants, while Post-Replay is +5.51, +5.65, +3.95, +6.23. For Gemini the corresponding Replay-Live values are +7.50, 0.00, -0.51, +5.62 and Post-Replay values are +13.43, +11.54, +2.49, +14.52. Post-Replay exceeds Replay-Live in all eight model-elicitation combinations of this small Replay cohort. We treat this as robustness evidence rather than as a new powered test.

H Held-Out Temperature Control

To test whether sampling numeric scores at temperature 0.7 was necessary for the observed state effects, we froze 10 unused English prompts, one from each original domain, and ran the exact Live-Replay-Post protocol for GPT-5.5 and Gemini at both $T = 0$ and $T = 0.7$. All 40 model-prompt-temperature Live/Post cells completed, as did every Replay sentence. Bootstrap intervals resample prompts with 20,000 deterministic replicates.

Table 8: Temperature control on 10 held-out prompts. Values are prompt means; intervals are prompt-bootstrap 95% CIs for Post-Live.

T	Model	$P - L$	95% CI	$R - L$	$P - R$
0.0	GPT-5.5	+5.17	[1.57, 9.88]	-1.97	+7.13
0.0	Gemini 3.1 Pro	+25.26	[16.80, 33.81]	-0.01	+25.27
0.7	GPT-5.5	+1.95	[-1.48, 6.20]	-3.72	+5.67
0.7	Gemini 3.1 Pro	+13.87	[8.33, 19.74]	-1.63	+15.50

At $T = 0$, Gemini-minus-GPT-5.5 Post-Live differs by +20.10 points (95% bootstrap CI [14.02, 26.54]). Mean within-output Live-Post Spearman is .730 for GPT-5.5 and .184 for Gemini at $T = 0$, so the qualitative ranking asymmetry also persists. Because temperature changes the generated prose trajectory as well as score-token sampling, the paired $T = 0$ versus $T = 0.7$ difference is not a clean causal estimate of temperature.

I Why Within-Output Ranking Is the Primary Local Metric

A pooled correlation asks whether a score on one prompt tends to correspond to a larger Post score than a different score on an unrelated prompt. A local controller usually asks a different question: among spans or siblings generated under one prompt, which candidate should be kept or edited? The within-output Spearman average and pairwise ordering agreement directly target this decision. The difference is material: GPT-OSS-120B has pooled English Live-Post Spearman .369 but mean within-output Spearman .640. Conversely, Qwen’s raw MAE is only 4.25 points but its mean within-output Spearman is .329. Reporting both axes avoids calling either model simply “well calibrated” or “poorly calibrated.”